

HOMEWORK 4

Track 2

Due date: October 31, 2023.

Reading assignment: Finish reading chapter 3 of the class notes (on class web-site) and start reading chapter 4. Read chapter 5 of the text book. Remember to include the self-assessment grades on the first sheet of your home work (see handout for details).

1. Show that for $p \geq 1$ and $p^{-1} + q^{-1} = 1$,

$$\|x\|_p = \max_{y \neq 0} \frac{|y^T x|}{\|y\|_q}, \quad x \in \mathbb{R}^n.$$

Hint: You can use Hölder's inequality for part of the proof.

2. Prove that for $v \in \mathbb{R}^n$

$$\|v\|_1 \|v\|_\infty \leq \frac{1 + \sqrt{n}}{2} \|v\|_2^2.$$

3. Suppose $u \in \mathbb{R}^m$ and $v \in \mathbb{R}^n$.

A) What is the rank of the matrix $u v^T$?

B) Show that $\|u v^T\|_F = \|u v^T\|_2 = \|u\|_2 \|v\|_2$.

C) Show that $\|u v^T\|_\infty = \|u\|_\infty \|v\|_1$.

4. Show that $\|A\|_2 \leq \|A\|_F$. *Note:* Please don't use the SVD.

5. Show that $\|A B\|_F \leq \|A\|_F \|B\|_F$.

6. Suppose $s \in \mathbb{R}^n$ and $s \neq 0$, and $E \in \mathbb{R}^{n \times n}$. Show that

$$\left\| E \left(I - \frac{s s^T}{s^T s} \right) \right\|_F^2 = \|E\|_F^2 - \frac{\|E s\|_2^2}{s^T s}.$$

Hint: Use the fact that $\|A\|_F^2 = \text{trace}(A^T A)$, and that $\text{trace}(A B) = \text{trace}(B A)$. If you use these hints, please prove them.

7. Show that for any real matrix A

$$\|A\|_2 = \sup_{\substack{x \neq 0 \\ y \neq 0}} \frac{|y^T A x|}{\|y\|_2 \|x\|_2}.$$

8. Show that $\|A\|_2 = \|A^T\|_2$ for any real matrix A .